

Efficient Bayesian Analysis of Longitudinal and Survival Data Using R-INLA

Joint modeling

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Pre-Conference Course

From separate models to a joint model

Survival

Differences in event risk between patients.

Longitudinal

Differences in marker levels and trajectories between patients.

Connecting the two

Can individual trajectories explain differences in event risk?

- ▶ Repeated measurements carry information about **patient heterogeneity** that may also be relevant to survival.
- ▶ The event can **end the measurements**: patients still observed later may be a selected group.

Joint modeling describes these related outcomes together.



ORIGINAL ARTICLE



Enfortumab Vedotin and Pembrolizumab in Untreated Advanced Urothelial Cancer

Authors: Thomas Powles, M.D., Begoña P. Valderrama, M.D., Shilpa Gupta, M.D., Jens Bedke, M.D., Eiji Kikuchi, M.D., Ph.D., Jean Hoffman-Censits, M.D., Gopa Iyer, M.D., ⁺¹⁷, for the EV-302 Trial Investigators* [Author Info & Affiliations](#)

Published March 6, 2024 | N Engl J Med 2024;390:875-888 | DOI: 10.1056/NEJMoa2312117 | **VOL. 390 NO. 10**

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RESULTS

A total of 886 patients underwent randomization: 442 to the enfortumab vedotin–pembrolizumab group and 444 to the chemotherapy group. As of August 8, 2023, the median duration of follow-up for survival was 17.2 months. Progression-free survival was longer in the enfortumab vedotin–pembrolizumab group than in the chemotherapy group (median, 12.5 months vs. 6.3 months; **hazard ratio for disease progression or death**, 0.45; 95% confidence interval [CI], 0.38 to 0.54; $P<0.001$), as was overall survival (median, 31.5 months vs. 16.1 months; **hazard ratio for death**, 0.47; 95% CI, 0.38 to 0.58; $P<0.001$). The median number of cycles was 12 (range, 1 to 46) in the enfortumab vedotin–pembrolizumab group and 6 (range, 1 to 6) in the chemotherapy group. Treatment-related adverse events of grade 3 or higher occurred in 55.9% of the patients in the enfortumab vedotin–pembrolizumab

Scientific rationale for covariate adjustment in RCTs



GUIDANCE DOCUMENT

Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products

MAY 2023

JOURNAL ARTICLE

Biased estimates of treatment effect in randomized experiments with nonlinear regressions and omitted covariates 🛒

[Get access >](#)

M. H. GAIL, S. WIEAND, S. PIANTADOSI

Biometrika, Volume 71, Issue 3, December 1984, Pages 431–444,

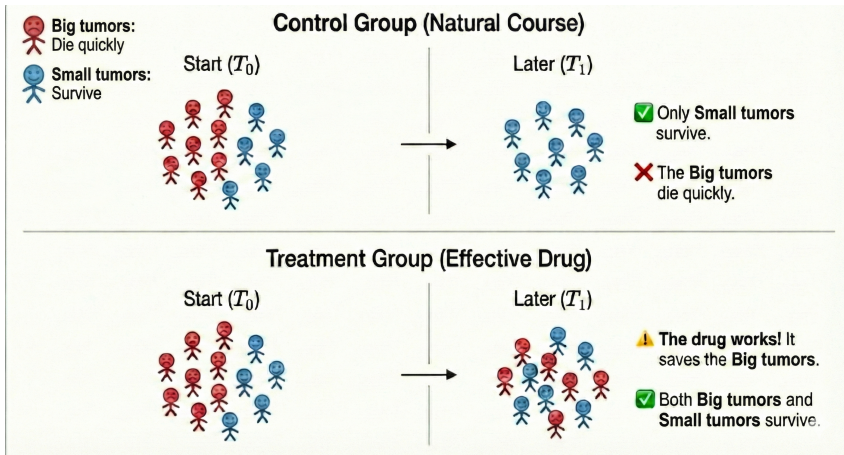
<https://doi.org/10.1093/biomet/71.3.431>

Published: 01 December 1984

[Article history ▼](#)

- ▶ **Precision:** Adjustment removes the “noise” of **patient heterogeneity** from the error term, **increasing statistical power**.
- ▶ **Balance:** It mathematically corrects for chance **imbalances in baseline characteristics** that occur despite randomization.
- ▶ **Effect Accuracy:** In Cox models, omitting prognostic factors mathematically **biases the Hazard Ratio toward 1.0**, diluting the drug’s apparent efficacy → **Attenuation bias**.

Attenuation bias - survival of the fittest



Unadjusted Cox model: Compares average risk between two groups.

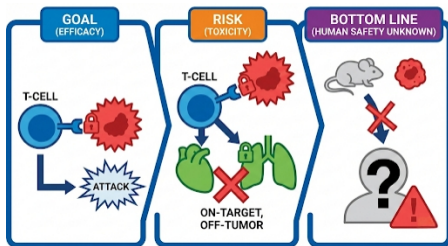
Adjusted Cox model: Compares conditional risk between two individuals with the same frailty level.

Tumor Size as a Dynamic Proxy for Disease State



A good treatment should simply reduce tumor size?

The Blind Spot of Tumor Shrinkage



- ▶ Reactivated T-cells identify and **attack cancer cells** based on a **specific antigen marker**.
- ▶ The same T-cells **attack healthy organs** (Heart, Lungs) because they share the **same antigen marker**.

If we only look at the tumor size, the drug looks efficient.

If we only look at survival, the drug looks like a failure.

→ What about a method that sees **both simultaneously**?

LONGITUDINAL SUBMODEL

$$Y_i(t) = \underbrace{\text{Fixed Effects}}_{\text{Average}(t)} + \underbrace{\text{Random Effects}}_{\text{Indiv. Deviation}(t)} + \underbrace{\epsilon}_{\text{Error}}$$

(Includes treatment effect!)

Joint modeling

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(Includes treatment effect!)

SURVIVAL SUBMODEL

Standard Cox

(Treats heterogeneity as statistical noise.
Adjusts only for baseline covariates)

$$h_i(t) = h_0(t) \cdot \exp(\beta \cdot \mathbf{Trt} + \gamma \cdot \mathbf{X}_{Baseline})$$

Joint modeling

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Joint model with Shared Random Effects (SRE)

(Adjusts for individual deviation from the population average)

$$h_i(t) = h_0(t) \cdot \exp(\cdots + \alpha \cdot \text{Deviation}_i(t))$$

Joint modeling

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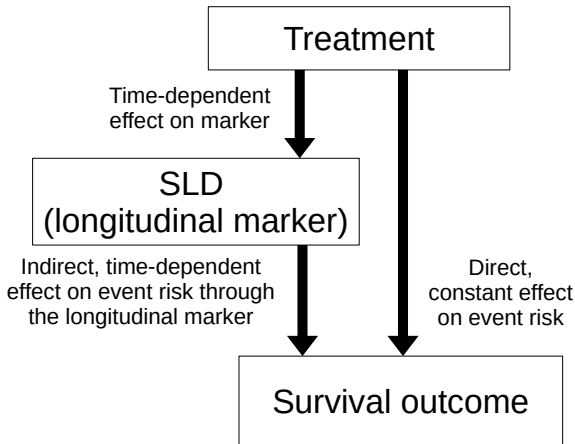
Joint model with Current Value (CV)

(Links risk to the longitudinal marker)

$$h_i(t) = h_0(t) \cdot \exp(\cdots + \alpha \cdot (\text{Average}(t) + \text{Deviation}_i(t)))$$

Joint Model with Current Value (JM CV)

$$HR(t) = \exp(\beta + \alpha \cdot \Delta_{\text{Treatment}}(t))$$



Joint model of longitudinal and survival data

- ▶ Links **longitudinal trajectories** with **time-to-event outcomes**.
- ▶ Accounts for **patient heterogeneity**, **measurement error** and **event-dependent dropout**.
- ▶ Enhances power by **using all data sources** simultaneously.
- ▶ Can **improve inference and prediction** compared with separate or two-stage analyses.

Joint modeling - Examples

A few examples of recent applications of joint models:

- ▶ Cancer tumor dynamics and the risk of death
- ▶ Analysis of CD4 lymphocytes counts and AIDS survival
- ▶ Prostate-specific antigen dynamics and the risk of cancer recurrence
- ▶ Dynamics of aortic gradient and aortic regurgitations and their relationship with the competing risks of death or reoperation in the field of cardiac surgery
- ▶ Cognitive markers's relationship with the time to onset of Alzheimer's disease
- ▶ Jointly modeling forest fires ignition, number of fires and the proportion of burned area in Portugal

Joint modeling - computational burden

Random effects used to fit the longitudinal marker need to be **integrated out** in the likelihood.

A **second level of numerical integration** is required when the risk function includes time-dependent variables.

$$\text{Survival likelihood} = \prod_{i=1}^N S_i(t) \lambda_i(t)^{[Eventindicator]}$$

where $S(t) = \exp(-\int_0^t \lambda(u) du)$ and $\lambda_i(t) = \lambda_0(t) \exp(X_i(t) \gamma)$

Association structure: Shared Random Effects

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha_0 b_{i0} + \alpha_1 b_{i1}\} \end{cases}$$

- ▶ Links risk to **patient-specific** latent traits (e.g., **independent** intercept and slope).
- ▶ **Computationally efficient**: avoids the complexity of time-dependent components while capturing heterogeneity.
- ▶ Asks if a patient's **baseline** marker value or **growth rate** is linked to survival outcomes.

Association structure: Individual deviation

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha(b_{i0} + b_{i1}t)\} \end{cases}$$

- ▶ Links risk to the **evolving individual deviation** from the population average at time t .
- ▶ Combines baseline and slope into **a single, time-varying deviation**.
- ▶ Isolates **residual heterogeneity**, without the effect of observed covariates like treatment.

Association structure: Current Value

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha \eta_i(t)\} \end{cases}$$

- ▶ Links risk to the **"true" underlying** biomarker level at time t .
- ▶ **Reduces bias**: filters out measurement error and noise from the observed data.
- ▶ Useful when **the marker is the biological driver**

Association structure: Current Slope

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp\{\gamma trt_i + \alpha \frac{d\eta_i(t)}{dt}\} \end{cases}$$

- ▶ Links risk to the current **rate of change** of the biomarker.
- ▶ Distinguishes a large stable tumor from a smaller but rapidly growing one.
- ▶ Asks if the speed of **disease progression** is an independent, prognostic factor of survival.

Association structure: Cumulative value (AUC)

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp \left\{ \gamma trt_i + \alpha \int_0^t \eta_i(s) ds \right\} \end{cases}$$

- ▶ Links risk to the **cumulative exposure** (Area Under the Curve) of the biomarker up to time t .
- ▶ Accounts for the **accumulated burden over time** rather than concomitant association.
- ▶ Asks if exposure (or **lagged exposure** within specific past windows) drives the event risk.

Association structure: Non-Linear Association

$$\begin{cases} \eta_i(t) = \beta_0 + b_{i0} + (\beta_1 + b_{i1})t + \beta_2 trt_i + \beta_3 trt_i t \\ \lambda_i(t) = \lambda_0(t) \exp \{ \gamma trt_i + f(\eta_i(t)) \cdot \eta_i(t) \} \end{cases}$$

- ▶ Links risk using a **flexible, data-driven function** (splines/random walks) rather than a restrictive linear assumption.
- ▶ Captures U-shaped, inverted-U, or threshold risk profiles **where linear models fail**.
- ▶ Asks if extreme values in both directions have **distinct** prognostic weight.

Dynamic Prediction

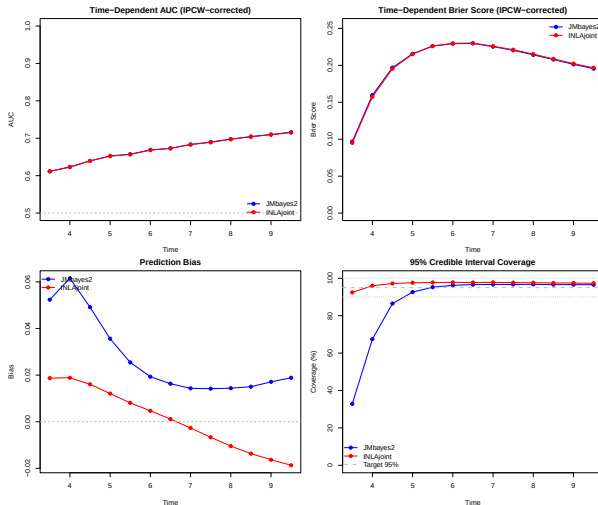
A standard Cox model gives a **static** survival curve at baseline.

A Joint Model enables **dynamic, real-time** risk updating.

- ▶ **Month 0:** Patient starts therapy. We predict their baseline survival curve.
- ▶ **Month 3:** Patient comes for a scan. Tumor has shrunk. We **update** the survival curve based on this new data.
- ▶ **Month 6:** Patient comes for another scan. Tumor is growing. The survival curve **dynamically drops**.

→ Joint models pave the way for **personalized, real-time medicine**.

Predictive accuracy metrics



Joint model with INLA

- ▶ Modular design that uses univariate models as **flexible building blocks**.
- ▶ Computation **speed**
- ▶ **Robust** estimation for highly parameterized models.
- ▶ **INLAjoint** simplifies thousands of lines of code into **concise syntax**.